A Quick Guide on CVXPY

CS4017: Introduction to Optimization

What is a solver?

Given an optimization problem, how to compute the optimal value / optimal variable? For example,

$$\begin{aligned} \underset{x,y}{\text{minimize}} \quad & (x-y)^2 \\ \text{subject to} \quad & x+y=1, \\ & x-y \geq 1. \end{aligned}$$

A solver helps you with that. All you have to do is write your problem into a **proper format**, and then input to the solver.

CVXPY

`CVX` was first developed by Michael Grant (student of Stephen Boyd) in 2006. Nowadays there are many solvers, with interfaces to various programming languages.
Here we mainly focus on `CVXPY`: `https://www.cvxpy.org/`, a Python package.

Install

CVXPY supports Python 3 on Linux, macOS, and Windows. You can use pip or conda for installation. You may want to isolate your installation in a virtualenv, or a conda environment.

Instructions

pip | conda | Install from source | Using Codespaces

(Windows only) Download the Visual Studio build tools for Python 3 (instructions).

(macOS only) Install the Xcode command line tools.

(optional) Create and activate a virtual environment.

Install CVXPY using pip:

```
pip install cvxpy
```

You can add solver names as "extras"; *pip* will then install the necessary additional Python packages.

```
pip install cvxpy[CBC,CVXOPT,GLOP,GLPK,GUROBI,MOSEK,PDLP,SCIP,XPRES
```

The first example

$$\begin{array}{ll} \underset{x,y}{\text{minimize}} & (x-y)^2 \\ \text{subject to} & x+y=1, \\ & x-y \geq 1. \end{array}$$

```python
import cvxpy as cp

# Create two scalar optimization variables.
x = cp.Variable()
y = cp.Variable()

# Create two constraints.
constraints = [x + y == 1,
               x - y >= 1]

# Form objective.
obj = cp.Minimize((x - y)**2)

# Form and solve problem.
prob = cp.Problem(obj, constraints)
prob.solve()  # Returns the optimal value.
print("status:", prob.status)
print("optimal value", prob.value)
print("optimal var", x.value, y.value)
```

```
status: optimal
optimal value 0.999999999761
optimal var 1.00000000001 -1.19961841702e-11
```

Constraints, Variables, and Parameters

`https://www.cvxpy.org/tutorial/intro/index.html`

1. Constraints: elementwise $==, <=, >=$; semidefinite $>>, <<$

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2. Variables: the decision variables, can be scalar, vector, or matrices (but not higher dimensions)

```
# A scalar variable.
a = cp.Variable()

# Vector variable with shape (5,).
x = cp.Variable(5)

# Matrix variable with shape (5, 1).
x = cp.Variable((5, 1))

# Matrix variable with shape (4, 7).
A = cp.Variable((4, 7))
```

Constraints, Variables, and Parameters

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1. Constraints: elementwise $==, <=, >=$; semidefinite $>>, <<$
2. Variables: the decision variables, can be scalar, vector, or matrices (but not higher dimensions)
3. Parameters: constants for a given problem instance: no need to re-write the code when the values change

```
# Positive scalar parameter.
m = cp.Parameter(nonneg=True)

# Column vector parameter with unknown sign (by default).
c = cp.Parameter(5)

# Matrix parameter with negative entries.
G = cp.Parameter((4, 7), nonpos=True)

# Assigns a constant value to G.
G.value = -numpy.ones((4, 7))
```

You can initialize a parameter with a value. The following code segments are equivalent:

```
# Create parameter, then assign value.
rho = cp.Parameter(nonneg=True)
rho.value = 2

# Initialize parameter with a value.
rho = cp.Parameter(nonneg=True, value=2)
```

Operators and functions

`https://www.cvxpy.org/tutorial/functions/index.html`

1. Operators: the same as Numpy: $+, -, *, /, @, .T, **$
2. Functions: most usual ones are already in-built

log_det(X)	$\log(\det(X))$	$X \in \mathbf{S}^n_+$	$\pm$ unknown	$\cap$ concave	None
log_sum_exp(X)	$\log\left(\sum_{ij} e^{X_{ij}}\right)$	$X \in \mathbf{R}^{m\times n}$	$\pm$ unknown	$\cup$ convex	$\nearrow$ incr.
matrix_frac(x, P)	$x^T P^{-1} x$	$x \in \mathbf{R}^n$ $P \in \mathbf{S}^n_{++}$	$+$ positive	$\cup$ convex	None
max(X)	$\max_{ij}\{X_{ij}\}$	$X \in \mathbf{R}^{m\times n}$	same as X	$\cup$ convex	$\nearrow$ incr.
mean(X)	$\frac{1}{mn}\sum_{ij}\{X_{ij}\}$	$X \in \mathbf{R}^{m\times n}$	same as X	$/$ affine	$\nearrow$ incr.
min(X)	$\min_{ij}\{X_{ij}\}$	$X \in \mathbf{R}^{m\times n}$	same as X	$\cap$ concave	$\nearrow$ incr.
mixed_norm(X, p, q)	$\left(\sum_k \left(\sum_l \lvert x_{k,l}\rvert^p\right)^{q/p}\right)^{1/q}$	$X \in \mathbf{R}^{n\times n}$	$+$ positive	$\cup$ convex	None
norm(x) norm(x, 2)	$\sqrt{\sum_i \lvert x_i\rvert^2}$	$X \in \mathbf{R}^n$	$+$ positive	$\cup$ convex	$\nearrow$ for $x_i \geq 0$ $\searrow$ for $x_i \leq 0$
norm(x, 1)	$\sum_i \lvert x_i\rvert$	$x \in \mathbf{R}^n$	$+$ positive	$\cup$ convex	$\nearrow$ for $x_i \geq 0$ $\searrow$ for $x_i \leq 0$
norm(x, "inf")	$\max_i\{\lvert x_i\rvert\}$	$x \in \mathbf{R}^n$	$+$ positive	$\cup$ convex	$\nearrow$ for $x_i \geq 0$ $\searrow$ for $x_i \leq 0$
norm(X, "fro")	$\sqrt{\sum_{ij} X_{ij}^2}$	$X \in \mathbf{R}^{m\times n}$	$+$ positive	$\cup$ convex	$\nearrow$ for $X_{ij} \geq 0$ $\searrow$ for $X_{ij} \leq 0$

Disciplined Convex Programming

Disciplined Convex Programming (DCP) helps CVXPY to check if the optimization problems are convex.
`https://www.cvxpy.org/tutorial/dcp/index.html`

1. Expressions: consist of variables, parameters (constants), operators, and functions.

```
import cvxpy as cp

# Create variables and parameters.
x, y = cp.Variable(), cp.Variable()
a, b = cp.Parameter(), cp.Parameter()

# Examples of CVXPY expressions.
3.69 + b/3
x - 4*a
sqrt(x) - minimum(y, x - a)
maximum(2.66 - sqrt(y), square(x + 2*y))
```

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1. Expressions: consist of variables, parameters (constants), operators, and functions.
2. Sign: flag each expression as *non-negative*, *non-positive*, *zero*, or *unknown*

```
x = cp.Variable()
a = cp.Parameter(nonpos=True)
c = numpy.array([1, -1])

print("sign of x:", x.sign)
print("sign of a:", a.sign)
print("sign of square(x):", cp.square(x).sign)
print("sign of c*a:", (c*a).sign)
```

```
sign of x: UNKNOWN
sign of a: NONPOSITIVE
sign of square(x): NONNEGATIVE
sign of c*a: UNKNOWN
```

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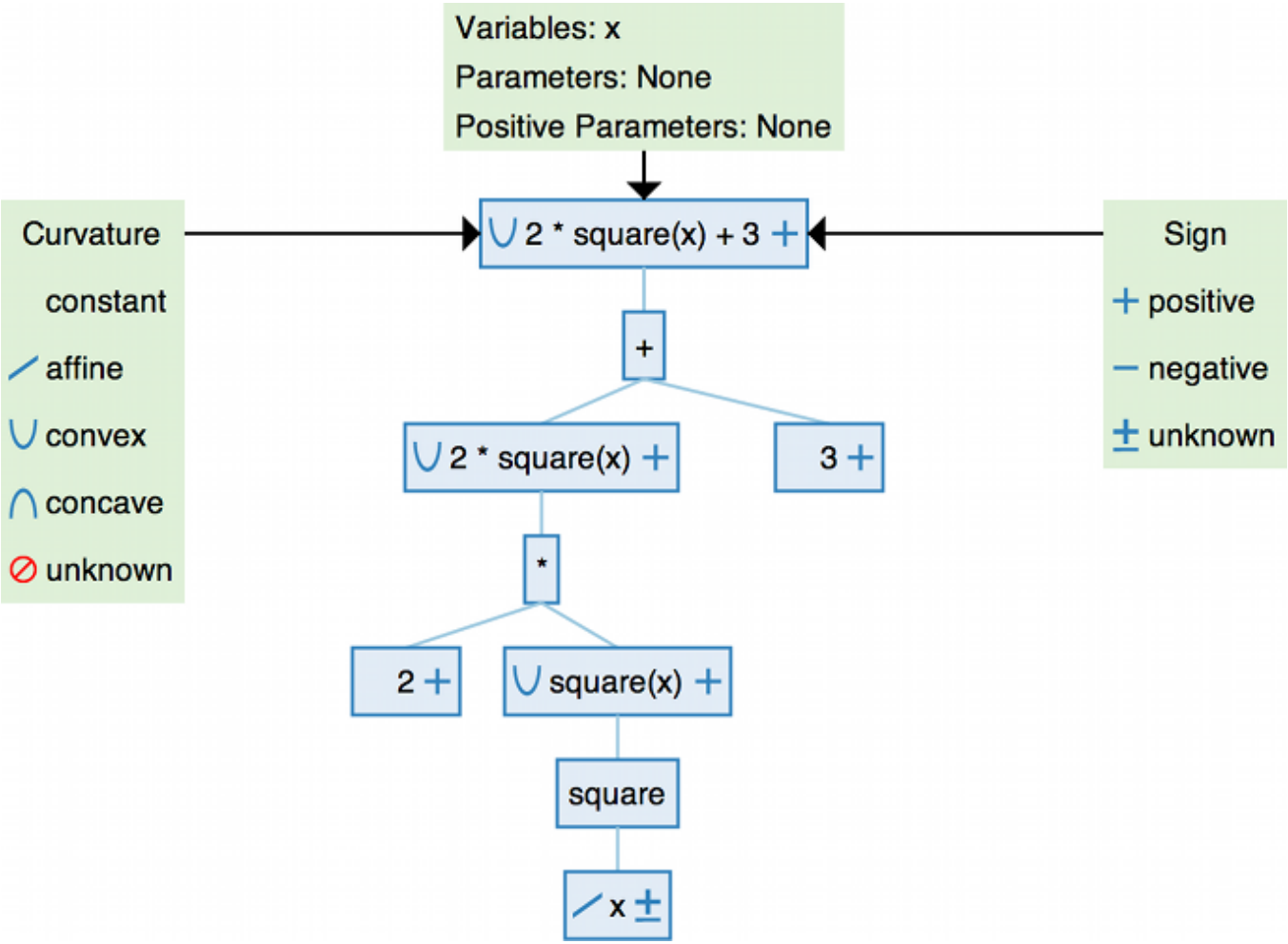
1. Expressions: consist of variables, parameters (constants), operators, and functions.
2. Sign: flag each expression as *non-negative*, *non-positive*, *zero*, or *unknown*
3. Curvature: apply composition theorem to decide if the expression is *convex*, *concave*, *affine*, or *unknown*. This is based on the sign, convexity, and monotonicity of each sub-expression.

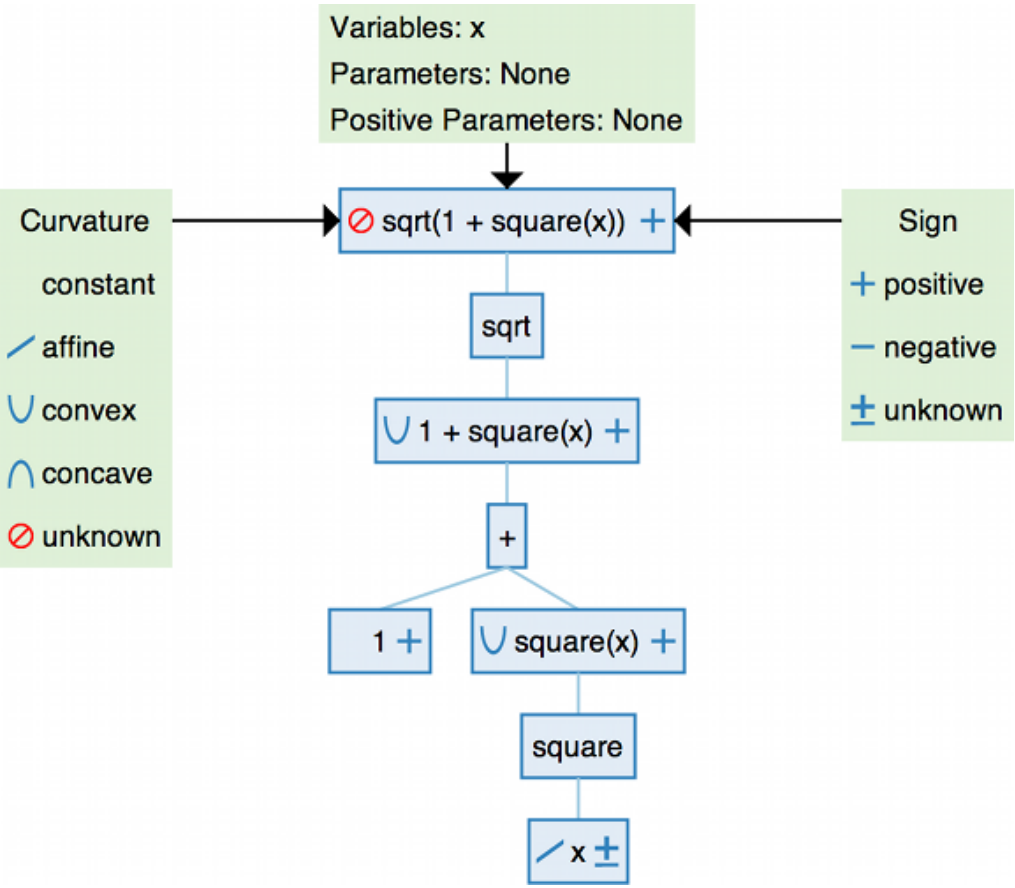
```
x = cp.Variable()
a = cp.Parameter(nonneg=True)

print("curvature of x:", x.curvature)
print("curvature of a:", a.curvature)
print("curvature of square(x):", cp.square(x).curvature)
print("curvature of sqrt(x):", cp.sqrt(x).curvature)
```

```
curvature of x: AFFINE
curvature of a: CONSTANT
curvature of square(x): CONVEX
curvature of sqrt(x): CONCAVE
```

DCP Examples



DCP Examples



Some expressions may not work. For example, sqrt() is *concave* and *increasing*, so it accepts only concave argument. By changing to another expression, it will be identified as convex.

```
print("sqrt(1 + square(x)) curvature:",
      cp.sqrt(1 + cp.square(x)).curvature)
print("norm(hstack([1, x]), 2) curvature:",
      cp.norm(cp.hstack([1, x]), 2).curvature)
```

```
sqrt(1 + square(x)) curvature: UNKNOWN
norm(hstack(1, x), 2) curvature: CONVEX
```

- Previously: $\sqrt{1+x^2}$
- Now: $\| [1, x] \|_2 = \left\| \begin{bmatrix} 1 \\ x \end{bmatrix} \right\|_2$

DCP Examples

A problem is constructed from an objective and a list of constraints. If a problem follows the DCP rules, it is guaranteed to be convex and solvable by CVXPY. The DCP rules require that the problem objective have one of two forms:

- Minimize(convex)
- Maximize(concave)

The only valid constraints under the DCP rules are

- affine == affine
- convex <= concave
- concave >= convex

Select a proper solver

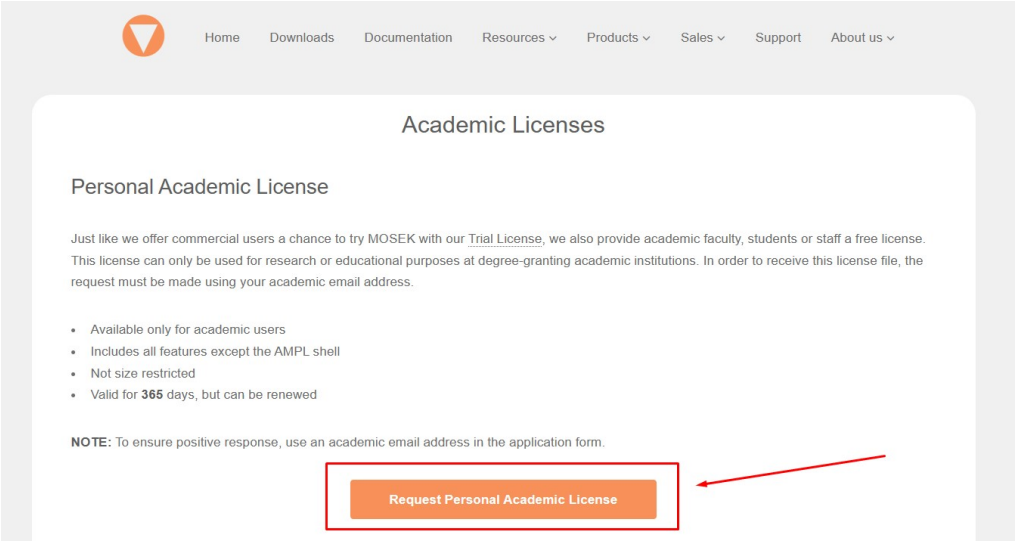
	LP	QP	SOCP	SDP	EXP	POW	MIP
CBC	X						X
CLARABEL	X	X	X	X	X	X	
COPT	X	X	X	X			X*
GLOP	X						
GLPK	X						
GLPK_MI	X						X
OSQP	X	X					
PIQP	X	X					
PROXQP	X	X					
PDLP	X						
CPLEX	X	X	X				X
NAG	X	X	X				
ECOS	X	X	X		X		
GUROBI	X	X	X				X
MOSEK	X	X	X	X	X	X	X**
CVXOPT	X	X	X	X			
SDPA ***	X	X	X	X			
SCS	X	X	X	X	X	X	
SCIP	X	X	X				X
XPRESS	X	X	X				X
SCIPY	X						X*

```
# Solve with MOSEK.
prob.solve(solver=cp.MOSEK)
print("optimal value with MOSEK:", prob.value)
```

See more in Link

Install MOSEK

1. Run ‘pip install cvxpy[MOSEK]‘
2. Goto `https://www.mosek.com/products/academic-licenses/`
   2.1 Request Personal Academic License, fill your information and receive an e-mail
   2.2 Accept the agreement and receive a ‘mosek.lic’, put it into proper place (follow the instruction in the e-mail)
3. Use MOSEK in your code via ‘prob.solve(solver=cp.MOSEK)‘



Examples: Quadratic Program

The standard form of quadratic program is:

$$\begin{array}{ll} \underset{x}{\text{minimize}} & (1/2)x^\top P x + q^\top x + r \\ \text{subject to} & Gx \preceq h, \\ & Ax = b, \end{array}$$

with $P \in \mathbb{S}^n_+$.

```python
# Import packages.
import cvxpy as cp
import numpy as np

# Generate a random non-trivial quadratic program.
m = 15
n = 10
p = 5
np.random.seed(1)
P = np.random.randn(n, n)
P = P.T @ P
q = np.random.randn(n)
G = np.random.randn(m, n)
h = G @ np.random.randn(n)
A = np.random.randn(p, n)
b = np.random.randn(p)

# Define and solve the CVXPY problem.
x = cp.Variable(n)
prob = cp.Problem(cp.Minimize((1/2)*cp.quad_form(x, P) + q.T @ x),
                 [G @ x <= h,
                  A @ x == b])
prob.solve()

# Print result.
print("\nThe optimal value is", prob.value)
print("A solution x is")
print(x.value)
print("A dual solution corresponding to the inequality constraints is")
print(prob.constraints[0].dual_value)
```

```
The optimal value is 86.89141585569918
A solution x is
[-1.68244521  0.29769913 -2.38772183 -2.79986015  1.18270433 -0.20911897
 -4.50993526  3.76683701 -0.45770675 -3.78589638]
A dual solution corresponding to the inequality constraints is
[ 0.          0.          0.          0.          0.         10.45538054
  0.          0.          0.         39.67365045  0.          0.
  0.         20.79927156  6.54115873]
```

Examples: Semidefinite Program

The standard form of semidefinite program is:

$$\begin{array}{ll} \underset{X \in \mathbb{S}^n}{\text{minimize}} & \mathbf{tr}(CX) \\ \text{subject to} & \mathbf{tr}(A_i X) = b_i,\ i = 1, \ldots, p, \\ & X \succeq 0, \end{array}$$

with $C, A_1, \ldots, A_p \in \mathbb{S}^n$.

```python
# Import packages.
import cvxpy as cp
import numpy as np

# Generate a random SDP.
n = 3
p = 3
np.random.seed(1)
C = np.random.randn(n, n)
A = []
b = []
for i in range(p):
    A.append(np.random.randn(n, n))
    b.append(np.random.randn())

# Define and solve the CVXPY problem.
# Create a symmetric matrix variable.
X = cp.Variable((n,n), symmetric=True)
# The operator >> denotes matrix inequality.
constraints = [X >> 0]
constraints += [
    cp.trace(A[i] @ X) == b[i] for i in range(p)
]
prob = cp.Problem(cp.Minimize(cp.trace(C @ X)),
                  constraints)
prob.solve()

# Print result.
print("The optimal value is", prob.value)
print("A solution X is")
print(X.value)
```

```
The optimal value is 2.654348003008652
A solution X is
[[ 1.6080571  -0.59770202 -0.69575904]
 [-0.59770202  0.22228637  0.24689205]
 [-0.69575904  0.24689205  1.39679396]]
```

Examples: Least Squares

The goal of least squares is to

$$\underset{x}{\text{minimize}} \quad \|Ax - b\|_2^2.$$

```python
# Import packages.
import cvxpy as cp
import numpy as np

# Generate data.
m = 20
n = 15
np.random.seed(1)
A = np.random.randn(m, n)
b = np.random.randn(m)

# Define and solve the CVXPY problem.
x = cp.Variable(n)
cost = cp.sum_squares(A @ x - b)
prob = cp.Problem(cp.Minimize(cost))
prob.solve()

# Print result.
print("\nThe optimal value is", prob.value)
print("The optimal x is")
print(x.value)
print("The norm of the residual is ", cp.norm(A @ x - b, p=2).value)
```

```
The optimal value is 7.005909828287484
The optimal x is
[ 0.17492418 -0.38102551  0.34732251  0.0173098  -0.0845784  -0.08134019
  0.293119    0.27019762  0.17493179 -0.23953449  0.64097935 -0.41633637
  0.12799688  0.1063942  -0.32158411]
The norm of the residual is  2.6468679280023557
```

Find out more

https://www.cvxpy.org/examples/index.html

Basic examples

- Least squares [.ipynb]
- Linear program [.ipynb]
- Quadratic program [.ipynb]
- Second-order cone program [.ipynb]
- Semidefinite program [.ipynb]
- Mixed-integer quadratic program [.ipynb]
- Control
- Portfolio optimization
- Worst-case risk analysis
- Model fitting
- Optimal advertising
- Total variation in-painting [.ipynb]

Disciplined geometric programming

- DGP fundamentals [.ipynb]
- Maximizing the volume of a box [.ipynb]
- Power control [.ipynb]
- Perron-Frobenius matrix completion [.ipynb]
- Rank-one nonnegative matrix factorization [.ipynb]

Disciplined quasiconvex programming

- Concave fractional function [.ipynb]
- Minimum-length least squares [.ipynb]
- Hypersonic shape design [.ipynb]

Derivatives

- Fundamentals [.ipynb]
- Queuing design [.ipynb]
- Structured prediction [.ipynb]

Machine learning

- Ridge regression [.ipynb]
- Lasso regression [.ipynb]
- Logistic regression [.ipynb]
- SVM classifier [.ipynb]
- Huber regression
- Quantile regression

Finance

- Portfolio optimization
- Cryptocurrency trading
- Entropic Portfolio Optimization
- Portfolio Optimization using SOC constraints
- Gini Mean Difference Portfolio Optimization
- Kurtosis Portfolio Optimization
- Relativistic Value at Risk Portfolio Optimization
- Approximate Kurtosis Portfolio Optimization

Advanced

- Object-oriented convex optimization [.ipynb]
- Consensus optimization [.ipynb]
- Method of multipliers [.ipynb]

Thank you

Any questions?